

Computer Networks

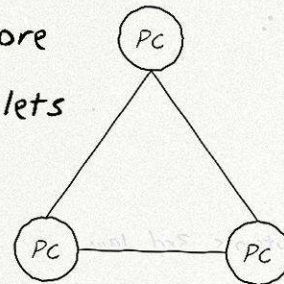
What is a Network?

A group of two or more similar things or people interconnected with each other is called a network.

- Social network , Mobile network
- Airlines, railway, bank, hospital networks

Computer Network

An interconnection among two or more computers or computing devices. It lets them ~~store~~ share data & resources.



Nodes & Packets

Node = any device in a network that can receive, create, store or send data.

Eg: server, laptop, modem, switch, router, printer, mobile phone.

Data is split into small chunks called packets, carried over wired (cables) or wireless (air) media.

Why use a Network?

Connecting computing devices gives many benefits in daily life.

Main advantages

① Resource sharing

One printer / storage used by many computers on the network.

② Information exchange

Email, websites, audio / video calls with many parties at the same time.

More benefits

- Communication is fast and cheap.
- Central data is easy to back up.
- Software / files shared from a server.
- Hotspot makes a small personal network.

Real-world note

A printer kept in one room can be used by every computer in the office through the network - no need to buy many.

Evolution of Networking

Networking began with ARPANET, a 1960s research project of the U.S. Dept. of Defence to link research institutions.

First message: UCLA $\leftrightarrow$ Stanford (SRI).

Timeline of milestones

- ① 1969 ARPANET working (UCLA-SRI)
- ② 1971 E-mail by Roy Tomlinson, '@'
- ③ 1974 term 'Internet' coined; Telenet
- ④ 1982 TCP/IP standard on ARPANET
- ⑤ 1983 Domain Name System introduced
- ⑥ 1990 Tim Berners-Lee: HTML, URL = WWW
- ⑦ 1997 First Wi-Fi (802.11) standard

Remember

WWW (1990) is ~~same~~ NOT the same as the Internet - Internet is much older (1969).

Types of Networks

Networks are classified by the geographical area covered and the data transfer rate.

Four main types

- ① PAN Personal Area Network
- ② LAN Local Area Network
- ③ MAN Metropolitan Area Network
- ④ WAN Wide Area Network

Range grows in this order

$PAN < LAN < MAN < WAN$

<- smallest
<- to largest

As area grows, speed (data rate) usually drops. LAN is fastest, WAN covers the globe but is slower per link.

The Internet is the largest WAN.

PAN and LAN

PAN (Personal Area Network)

A few personal devices connected within about 10 metres.

- Wired PAN : phone to laptop by USB.
- * Wireless PAN (WPAN) : Bluetooth link.

LAN (Local Area Network)

Connects devices in a limited area : a room, floor, office, lab or campus.

- Wires: Ethernet cable, fibre, or Wi-Fi.
- Secure - only authentic users join.
- Can extend up to about 1 km.
- Speed 10 Mbps (Ethernet) to 1000 Mbps (Gigabit Ethernet).

Ethernet

A set of rules that decides how computers and devices connect through cables in a LAN. Mbps = Megabits per second.

$1 \text{ Gbps} = 1000 \text{ Mbps}$

MAN and WAN

MAN (Metropolitan Area Network)

An extended LAN covering a city or town.

- Speed less than LAN, still in Mbps.
- Eg: cable TV, cable broadband internet.
- Reaches about 30 - 40 km.
- Often many LANs joined together.

WAN (Wide Area Network)

Connects LANs and MANs across cities, countries or continents.

- Used by big business / govt branches.
- The Internet is the largest WAN.

LAN vs WAN

LAN	WAN
Small area	Very large area
High speed	Lower speed
Owned privately	Shared / public

Network Devices: Modem

Devices like Modem, Hub, Switch, Repeater, Router, Gateway help configure networks.

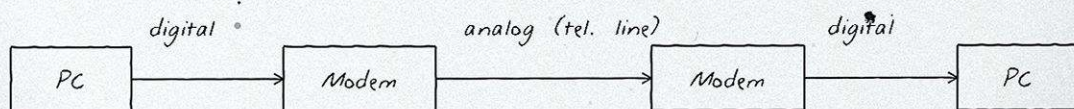
Modem = MODulator DEModulator

Converts between analog signals and digital bits (0s and 1s).

- Sender's modem = MODULATOR
 - digital $\rightarrow$ analog
- Receiver's modem = DEMODULATOR
 - analog $\rightarrow$ digital

Modem flow

*



Computers store data as bits, but the line carries an analog signal. So a modem sits at both ends to translate the signal.

NIC, RJ45 & Repeater

Ethernet Card (NIC)

Network Interface Card - a network adapter for a wired network. It is the interface between a computer and the network.

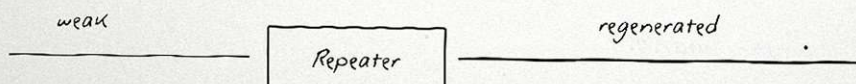
- Circuit board on the motherboard.
- Speed 10 Mbps to 1 Gbps.
- Each NIC has a unique MAC address.

RJ45 (Registered Jack-45)

An 8-pin connector used only with Ethernet cables. The small plastic plug at the end of every network cable.

Repeater

Signals weaken after about 100 m on a cable. A repeater is an analog device that ~~copies~~ regenerates the weak signal and puts it back on the cable.



Hub and Switch

Hub

Connects devices through wires. Data arriving on one line is sent out on ALL other lines (broadcast).

- Drawback: if two devices send at once, the data collide.

Switch

Central device of a LAN. It reads the destination address from each packet and sends it only to the selected device.

- Forwards many packets at the same time.
- Drops noisy / corrupt signals, asks to
* resend.

Hub vs Switch

Hub -> sends to ALL (collisions).

Switch -> sends to ONE (smart, no clash).

Switch is smarter than a Hub

Router and Gateway

Router

Receives data, analyses^{*} it and transmits it to OTHER networks. Connects a LAN to the Internet.

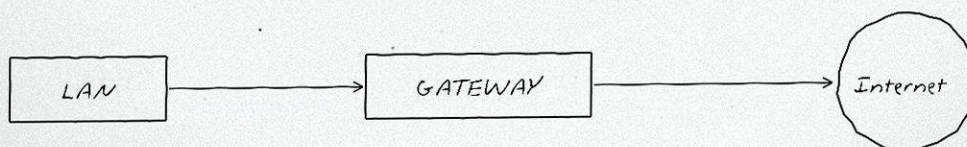
- Can repack big packets into small ones for a different network.
- Wireless router gives Wi-Fi access.
- Home Wi-Fi router = router + modem.

*

Gateway

A 'gate' between an organisation's network and the outside Internet.

- Entry & exit point - all data passes it.
- Routes to a foreign network's node.
- Often the firewall is built into it.



Usually a router is set up as the gateway.

Network Topologies

Topology = the arrangement of computers and peripherals in a network.

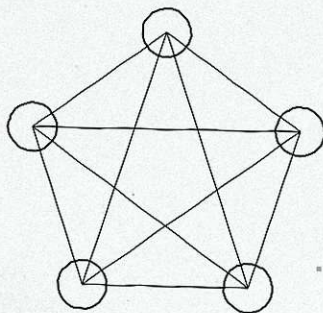
Mesh , Ring , Bus , Star , Tree

*

1. Mesh Topology

Each device is connected to EVERY other device in the network.

- Handles heavy traffic (many at once).
- Reliable - one node down, others run.
- Secure - each cable carries own data.
- But: complex wiring, high cabling cost.



wires for mesh: $n \text{ nodes} \Rightarrow \text{wires} = n(n-1)/2$

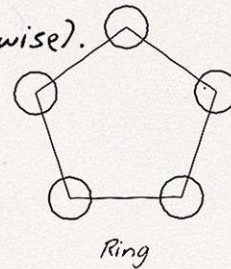
Ring & Bus Topology

2. Ring Topology

Each node connects to two others, one on each side, forming a ring.

- Link is unidirectional - data flows one way (clockwise or anti-clockwise).

*



3. Bus Topology

All devices connect to a single backbone wire called the BUS.

- Data travels both ways along the bus.
- Cheaper and easy to maintain.
- Ring & Bus: less secure, less reliable.



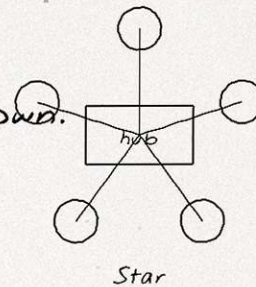
Bus topology (shared backbone)

Star & Tree Topology

4. Star Topology

Every device connects to a central node (a hub or switch).

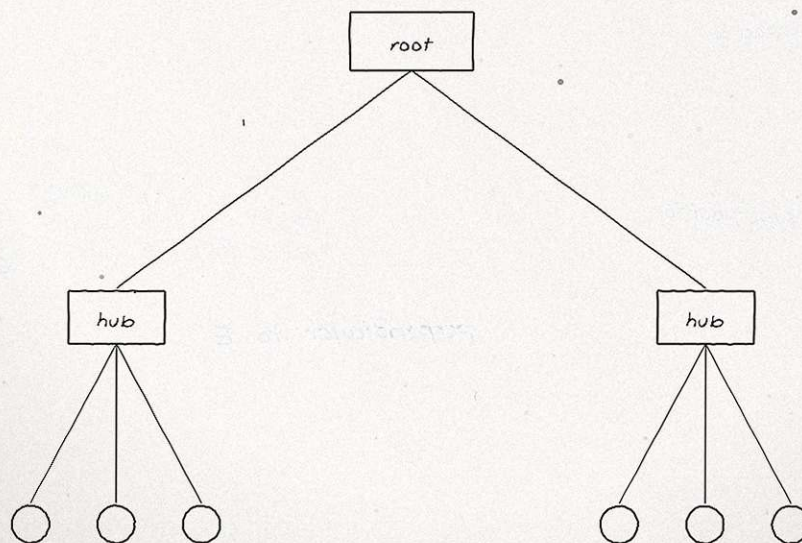
- Fast & efficient - direct to centre.
- One device fails $\rightarrow$ rest are fine.
- Central device fails $\rightarrow$ whole net down.



5. Tree / Hybrid Topology

A hierarchical topology with branches; each branch can be star, ring or bus.

- Common in WANs joining many LANs.
- Data goes via the central device first.



Tree = stars hung on a backbone

Comparing Topologies

Bus vs Tree (Exam Q8)

- Bus: ONE backbone, all nodes on it,
- Flat (no hierarchy).
- Tree: hierarchical, many branches, each
- branch may be star / ring / bus.
- Tree joins several topologies; bus is
- a single shared line.

Quick comparison

- ① Mesh most reliable, costliest wiring
- ② Ring one-way flow, node down = break *
- ③ Bus cheap, single backbone wire *
- ④ Star central hub, easy to add nodes
- ⑤ Tree hierarchy of stars / buses

Identify topology (Exam Q9)

- a) single cable to all → BUS
- b) central switch, own cables → STAR

MAC Address

Each node must be uniquely identified so a device knows the sender & receiver.

MAC = Media Access Control

- Also called physical / hardware address.
- A unique value tied to the NIC.
- Engraved at manufacture - PERMANENT,
- cannot be changed.
- Physically identifies a machine.

Format

12-digit hexadecimal = 48 bits.

- First 6 digits (24 bits) = maker ID (OUI).
- Last 6 digits (24 bits) = card serial no.

MAC : 48 bits = 6 + 6 hex digits

*

Significance (Exam Q13)

It is the permanent, globally unique tag that lets the network physically find an exact machine - no two NICs share one.

IP Address

IP = Internet Protocol address

A unique address to identify each node that uses the Internet Protocol.

- Know the IP $\rightarrow$ reach that computer.
- CAN change when a node moves networks.

IPv4 and IPv6 *

- IPv4 : 32-bit, 4 numbers (0-255), dots.

Eg: 192.168.0.178

- Gives under 4.3 billion addresses.
- IPv6 : 128-bit, 8-hex groups, colons.

Eg: 2001:CDBA:::3257:9652

MAC vs IP (Exam Q14)

MAC	IP
Physical addr	Logical addr
Permanent	Can change
Set by maker	Set by network

Internet, Web & IoT

The Internet

Global network of computing devices :
desktops, laptops, servers, phones,
printers, routers, switches, gateways.

- Smart TV, AC, fridge can join too (IoT).
- Modem → local ISP → regional →
national → Internet backbone.

ISP

Internet Service Provider - an organisation
that provides access to the Internet.



Internet of Things (IoT)

Everyday smart devices connected to the
network - drones, vehicles, door locks,
cameras. (IoT & WoT studied in class XI.)

World Wide Web (WWW)

An ocean of information stored as trillions of interlinked web pages & resources, shared / accessed through the Internet.

Invented by Sir Tim Berners-Lee in 1990 using three core technologies:

Three web technologies

① HTML HyperText Markup Language

designs standard web pages.

② URI / URL Uniform Resource

Identifier/Locator = unique web address of a resource.

③ HTTP HyperText Transfer Protocol

rules to retrieve linked pages;

secure version = HTTPS.

Internet vs Web (Exam Q16)

Internet = the huge network of computers.

Web = the linked web pages stored on those computers, reached via Internet.

Domain Name System

URL anatomy

A URL is more than the domain name.

`http://www.ncert.nic.in/textbook/.htm`

domain

path

Why DNS?

Every server has an IP address, but numbers are hard to remember. So each server gets a domain (host) name mapped to its IP.

- Like a phonebook: name $\leftrightarrow$ number.

Example mapping

ncert.nic.in $\rightarrow$ 164.100.60.233

wikipedia.org $\rightarrow$ 198.35.26.96

DNS = Domain Name System

The system that maps domain names to IP addresses. Converting a name to its IP is called ~~name lookup~~ domain name resolution.

It is done by a special server called the DNS server.

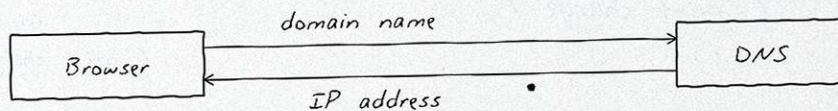
DNS Server & Lookup

DNS server

Keeps a database of domain names and their matching IP addresses.

- Servers are placed in a hierarchy.
- Top level = 13 root servers (A to M).
- A server gives the IP, or points to another DNS server to ask.

How resolution works



- ① Type URL -> HTTP asks DNS server.
- ② DNS returns the matching IP.
- ③ HTTP fetches the page from that IP and loads it in the browser.

IANA keeps the list of root servers.

Exercise: Key Answers

Q1. Expand

- ARPANET = Advanced Research Projects Agency Network
- MAC = Media Access Control *
- ISP = Internet Service Provider
- URI = Uniform Resource Identifier

Q5. Common networking devices

Modem, Hub, Switch, Repeater, Router, Gateway, Ethernet card (NIC), RJ45.

Q6. Two states, transfer info

Different states are far apart, so they need a WAN (Wide Area Network).

Q10. Modem - why used?

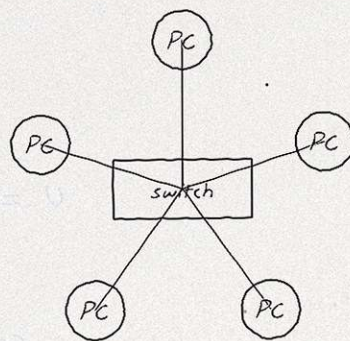
Converts digital bits to analog (and back) so data can travel over the line / internet.

Q15. DNS

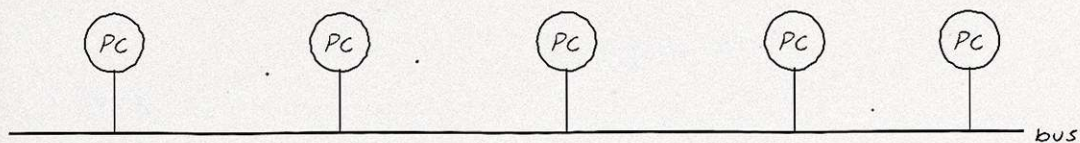
DNS maps domain names to IP addresses. A DNS server stores this database and resolves a name to its IP on request.

Q12: Layouts (5 PCs)

Star topology



Bus topology



Star: each PC has its own cable to the switch. Bus: all 5 share one backbone.

Summary - I

- Network = interconnected computing devices
- sharing data & resources.
- Node = device that sends/receives data.
- Data travels as small packets.
- Types by area: PAN < LAN < MAN < WAN.
- LAN: limited area, fast, secure, up to 1 km.
- MAN: city-wide (30-40 km), eg cable TV.
- WAN: countries/continents; Internet = largest WAN.
- Modem: digital <-> analog signal.
- NIC: wired adapter, holds the MAC address.
- Repeater: regenerates a weak signal. *
- Hub: sends to all (collisions).
- Switch: sends to the right device only.
- Router: links a LAN to other networks.
- Gateway: entry/exit point of a network.

Topologies & addressing on the next page.

Summary - II

- Topology = arrangement of nodes.
- Mesh: all-to-all, reliable, costly.
- Ring: *each node to two, one-way flow.
- Bus: one shared backbone, cheap.
- Star: central hub/switch, easy to manage.
- Tree: hierarchy of stars/buses (WAN). *
- MAC: 48-bit, permanent, physical address.
- IP: logical address, can change; v4=32-bit,
v6=128-bit.
- Internet = global network of devices.
- WWW (1990, Berners-Lee) = linked web pages.
- Web tech: HTML, URI/URL, HTTP(S).
- DNS: maps domain name $\rightarrow$ IP address.
- DNS server resolves the name on request;
- 13 root servers at the top.

These points cover the whole chapter.

Common Mistakes

① Internet vs Web

Internet = network; Web = pages on it.

② Hub vs Switch

Hub broadcasts to all; switch is selective (one device).

③ MAC vs IP

MAC permanent/physical; IP logical and changeable.

④ Router vs Gateway

Router links networks; gateway is the single entry/exit gate.

⑤ Modem direction

Modulate = digital → analog (send);

demodulate = analog → digital (recv).

Tip

Draw the topology while answering - it earns marks and avoids confusion.

Quick Recall

Remember these

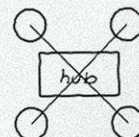
PAN < LAN < MAN < WAN

Mesh wires = $n(n-1)/2$

MAC = 48 bit , IPV4 = 32 bit

Mnemonics

- Topologies: 'My Ring Bus Stars Trees'
- = Mesh, Ring, Bus, Star, Tree.
- Modem = MODulate + DEModulate.
- Switch is Smart, Hub is dumb (broadcast).
- DNS = Domain Name $\rightarrow$ IP, the 'phonebook'.



Prev: Searching Next: Database Concepts